

Unit 1: Introduction to Machine Learning (Scientific View)

Machine Learning (ML) is a scientific discipline that focuses on how computers can learn patterns and make decisions from data without being explicitly programmed. It is a subset of Artificial Intelligence (AI), with core applications in classification, prediction, optimization, and control.

ML involves the development of algorithms that can generalize from observed data and improve their performance as more data becomes available.

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Types of Machine Learning

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Supervised Learning
Example: Spam detection

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Unsupervised Learning
Example: Customer segmentation

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Semi-supervised Learning
Example: Image recognition

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Reinforcement Learning
Example: Game AI

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 - Financial modeling and risk analysis

ML is fundamentally rooted in statistics, linear algebra, optimization, and probability theory. Algorithms are evaluated based on generalization error, bias-variance tradeoff, and computational efficiency.